

CSE110 Fall 2025 Lab Exam - 02 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;
public class TemperatureScore {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n;
        System.out.print("Enter n: ");
        n = sc.nextInt();
        int temperatureScore = 0;
        for (int i = 1; i <= n; i++) {
            int temp;
            System.out.print("Enter temperature " + i + ": ");
            temp = sc.nextInt();

            if (temp < 50 || temp > 150) {
                System.out.println("Invalid temperature, "+temp+".");
                i -= 1; // re-prompt for same reading
            } else {
                if (temp % 5 == 0 || temp % 4 == 0) {
                    temperatureScore += temp;
                } else {
                    temperatureScore -= temp;
                }
            }
        }
        int tempScoreCopy = temperatureScore;
        int evenDigitCount = 0;
        if (tempScoreCopy < 0) {
            tempScoreCopy *= -1;
        }
        while (tempScoreCopy != 0) {
            int digit = (tempScoreCopy % 10);
            if (digit % 2 == 0) {
                evenDigitCount++;
            }
            tempScoreCopy /= 10;
        }
        System.out.println("\nFinal Temperature Score: " + temperatureScore);
        System.out.println("Number of even digits: " + evenDigitCount);
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
<u>1</u>	Imports Scanner & takes input n & temperature	1.5
<u>2</u>	Runs the first loop n times.	1
<u>3</u>	Checks whether the temperature is within the range 50–150 and reprompts if it is not.	1.5
<u>4</u>	Initializes the variable temperatureScore and updates it according to the correct conditions.	2.5
<u>5</u>	Count the number of even digits of the temperature score & check for negative numbers.	2.5
<u>6</u>	Prints the Temperature Score and digit count.	1

[SET-B]

SOLUTION

```
import java.util.Scanner;
public class RainfallScore {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n;
        System.out.print("Enter K: ");
        int K = sc.nextInt();
        int rainfallScore = 0;
        for (int i = 1; i <= K; i++) {
            int rainfall;
            System.out.print("Enter rainfall " + i + ": ");
            rainfall = sc.nextInt();

            if (rainfall < 0 || rainfall > 200) {
                System.out.println("Invalid rainfall value, "+rainfall+".");
                i--; // re-prompt for same reading
            } else {
                if (rainfall % 7 == 0 || rainfall%3==0) {
                    rainfallScore += rainfall;
                } else {
                    rainfallScore -= rainfall;
                }
            }
        }
        int scoreCopy = rainfallScore;
        int oddDigitCount = 0;
        if (scoreCopy<0){
            scoreCopy*=-1;
        }
        while (scoreCopy != 0) {
            int digit = (scoreCopy % 10);
            if (digit % 2 != 0) {
                oddDigitCount++;
            }
            scoreCopy /= 10;
        }
        System.out.println("\nFinal Rainfall Score: " + rainfallScore);
        System.out.println("Number of odd digits: " + oddDigitCount);
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
<u>1</u>	Imports Scanner & takes input n & rainfall values	1.5
<u>2</u>	Runs the first loop n times.	1
<u>3</u>	Checks whether the rainfall is within the range 0–200 and reprompts if it is not.	1.5
<u>4</u>	Initializes the variable rainfallScore and updates it according to the correct conditions.	2.5
<u>5</u>	Count the number of odd digits of the rainfall score & check for negative numbers.	2.5
<u>6</u>	Prints the Rainfall Score and digit count.	1